

Custom Logic with `.apply()` and `.map()`

- **The Power of Customization:** When built-in Pandas functions aren't enough, we write our own.
- **`.map()`:** Used for element-wise transformations on a Series (e.g., changing 'M' to 'Male').
- **`.apply()`:** The "Swiss Army Knife." It can run a function across an entire row or column, allowing for complex logic that depends on multiple variables simultaneously.

[1]:

	CustomerID	Name	City	Product	Amount	Tier
0	201	Nina	Chennai	Book	6705	Gold
1	202	Nina	Kolkata	Mobile	17572	Silver
2	203	Tina	Delhi	Mobile	5964	Gold
3	204	Nina	Kolkata	Book	10143	Bronze
4	205	Kavya	Pune	Mobile	2088	Gold

Before transformation

[5]:

```
df["Name"] = df["Name"].str.upper()
df["City"] = df["City"].str.strip()
df["Product"] = df["Product"].str.replace("Book", "Notebook")
df["Name"] = df["Name"].str.len()

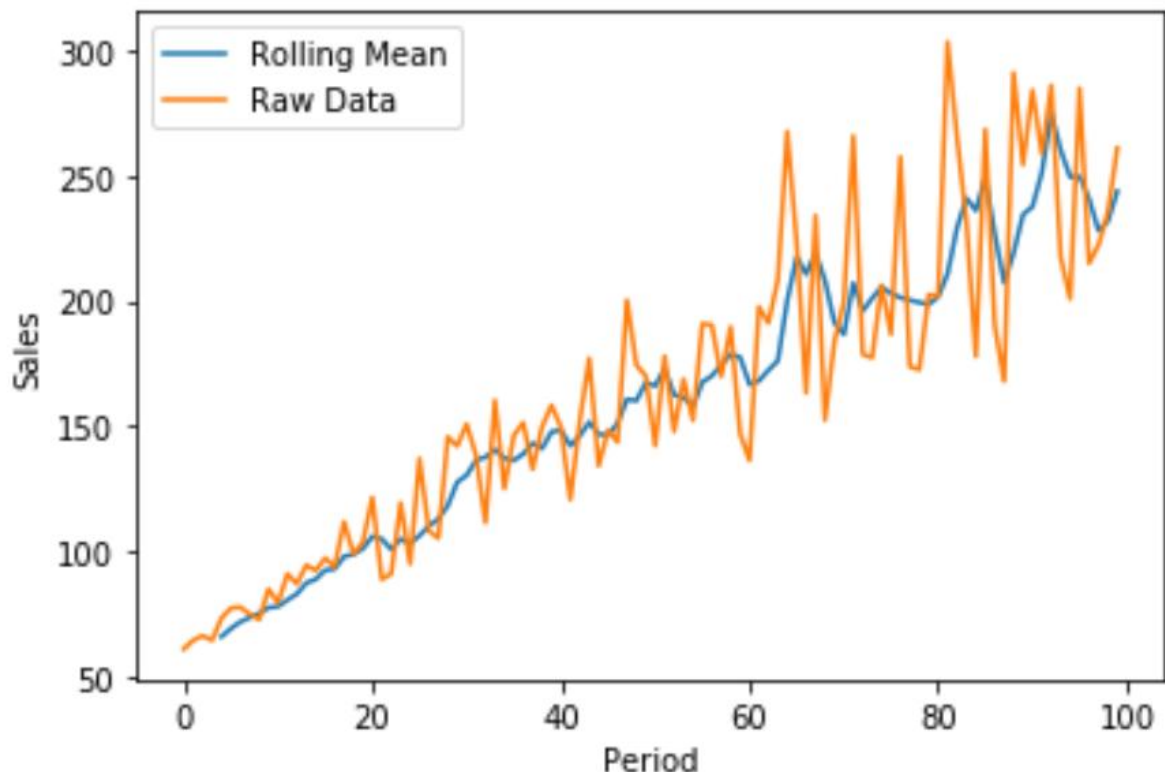
df.head()
```

[5]:

	CustomerID	Name	City	Product	Amount	Tier
0	201	NINA	Chennai	Notebook	6705	Gold
1	202	NINA	Kolkata	Mobile	17572	Silver
2	203	TINA	Delhi	Mobile	5964	Gold
3	204	NINA	Kolkata	Notebook	10143	Bronze
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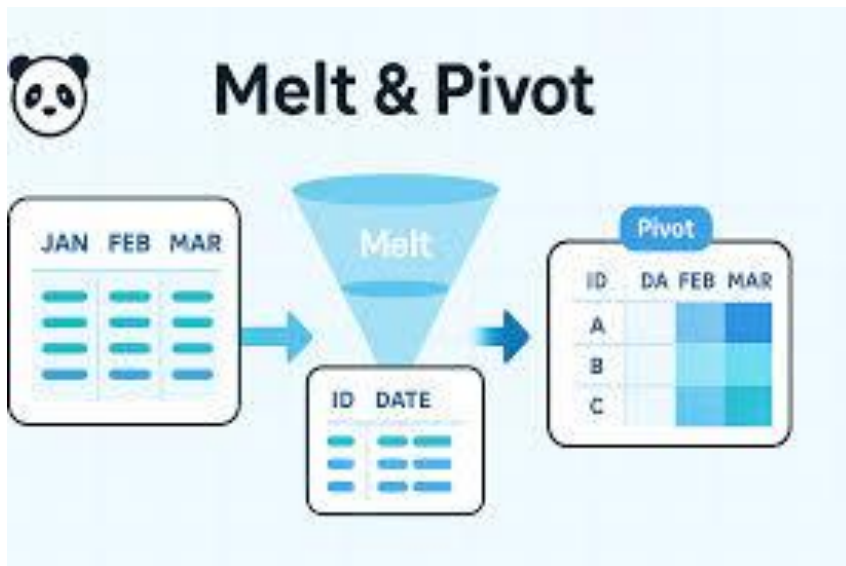
After transformation

Window Functions (Rolling Statistics)



- **Analyzing Trends:** In real-world data (like stock prices or weather), single data points are often "noisy."
- **Rolling Windows:** Calculating metrics over a moving subset of data (e.g., a 7-day average).
- **The Goal:** Smoothing out short-term fluctuations to see the long-term trend or "signal" in the data.

Reshaping Data: Pivoting & Melting



- **Wide vs. Long Format:**
 - **Wide:** Easy for humans to read (e.g., a column for every month).
 - **Long:** Necessary for computer analysis and visualization (e.g., one 'Date' column and one 'Value' column).
- **Pivoting:** Turning "Long" data into "Wide" (Summarizing).
- **Melting:** Turning "Wide" data into "Long" (Un-pivoting for analysis).

Method Chaining (The Professional Standard)



- **Code Quality:** Avoid creating "Intermediate Variables" (like `df1`, `df2`, `df3`) that clutter your computer's memory.
- **The Pipeline:** Linking operations together in a single, readable block of code.
- **Readability:** Makes the data transformation steps look like a clear recipe that anyone on your team can follow.